

Junxiang Huang

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Google Scholar



Education

Peking University

PhD Student in Quantum Information and Quantum Computing

Sep. 2022 – Expected Jun. 2027

Beijing, China

Peking University

B.S. in Physics and B.S. in Computer Science

Sep. 2018 – Jul. 2022

Beijing, China

Research Experience

Xiao Yuan's Group

Center on Frontiers of Computing Studies, Peking University

Graduate Researcher in Quantum Information and Quantum Computing

Apr. 2021 – Present

Beijing, China

- Developed resource-efficient quantum algorithms and theoretical tools for many-body simulation and information extraction, spanning variational, tensor-network-assisted, and early fault-tolerant approaches.
- Designed qubit-, measurement-, and sample-efficient protocols for estimating local observables and nonlinear quantum-state properties, including partial-transpose moments.
- Current projects include response-transform and counterdiabatic algorithms, fault-tolerant resource estimation, and learning and simulation of open quantum dynamics.

Quantum Information Theory Group at Imperial College London

Electrical and Electronic Engineering Department, Imperial College London

Visiting Research Student

Jul. – Aug. 2023

London, UK

- Studied applications of perturbative quantum simulation to quantum information and many-body dynamics.

Research Interests

- Resource-efficient quantum algorithms for many-body simulation, eigenstate and dynamical problems, with particular emphasis on the early fault-tolerant regime.
- QEC/QEM-aware estimation protocols, algorithm–architecture resource analysis, and hybrid quantum–classical methods for quantum simulation.

Publications & Preprints

[9] Y. Tang*, Z. Huo*, **J. Huang***, et al. Low-cost algorithm-to-execution framework for surface-code quantum computing. arXiv:2609.10965, 2026.

*Equal contribution.

[8] J. Cai*, **J. Huang***, F. Otto*, Y. Li*, et al. Generalized Mermin Inequalities for Benchmarking Large-Scale GHZ States. arXiv:2607.23574, 2026.

*Equal contribution.

[7] J. Sun*, B. Zhou*, J. Xu*, Y. Yao*, Z. Du*, Z. Zhang*, Y. Gu*, **J. Huang**, et al. Quantum-classical crossover in fault-tolerant quantum dynamics simulation. arXiv:2607.16116, 2026.

*Equal contribution.

[6] **J. Huang**, X. Wang, X. Yuan, Y. Zhang. Simultaneous Estimation of Partial-Transpose Moments with Active Memory Independent of the Moment Order. arXiv:2606.14204, 2026.

[5] Z. Wang*, **J. Huang***, R. Ye*, et al. A Review of Variational Quantum Algorithms: Insights into Fault-Tolerant Quantum Computing. arXiv:2604.07909, 2026.

*Equal contribution.

[4] Q.-M. Ding, J. Peng, **J. Huang**, et al. Obtaining Accurate Ground-State Properties on Near-term Quantum Devices. Phys. Rev. Applied 25, 054057 (2026).

[3] **J. Huang**, Y. Tang, X. Yuan. Estimating Local Observables via Cluster-Level Light-Cone Decomposition. Phys. Rev. A 113, 032414 (2026).

[2] T. Jiang*, J. Cai*, **J. Huang***, N. Zhou*, Y. Zhang, et al. One- and two-dimensional cluster states for topological phase simulation and measurement-based quantum computation. Nat. Phys. 22, 430–438 (2026).

*Equal contribution.

[1] **J. Huang***, W. He*, Y. Zhang, et al. Tensor-Network-Assisted Variational Quantum Algorithm. Phys. Rev. A 108, 052407 (2023).

*Equal contribution.

Invited Talks

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems

Shenzhen International Quantum Academy

Jul. 2025

Referee Service

Journal referee service: Physical Review Letters, Physical Review A, npj Quantum Information, Quantum

Conference referee service: AQIS, TQC

Awards & Honors

Award for Scientific Research, 2024–2025 Academic Year

Peking University

Dec. 2025

Special Prize, China Youth Science and Technology Innovation Challenge Track

19th “Challenge Cup” National College Students’ Extracurricular Academic S&T Works Competition

Nov. 2025

3rd Place Award, Student Poster Presentation

AQIS 2023 Seoul

Sep. 2023

Peking University Teaching Assistant Scholarship

Peking University

Sep. 2022 – Present

Award for Scientific Research, 2019–2020 Academic Year

Peking University

Dec. 2020

Award for Academic Excellence, 2018–2019 Academic Year

Peking University

Dec. 2019

First Prize, National Final Round; Selected for the National Training Team

34th Chinese Physics Olympiad

Nov. 2017

Selected Funded Projects

Beijing Natural Science Foundation Youth Project, **Participant** (2026–present): Quantum simulation and probing algorithms for dynamical quantum phase transitions.

Beijing Natural Science Foundation Interdisciplinary Key Project, **Participant** (2025–present): Mathematical foundations, algorithms, and applications for practical quantum information processing.

Patents

- Method and Apparatus for Quantum Variational Molecular Dynamics Simulation (CN119811513B)
- Hybrid Computing Method and Apparatus Based on Neural Network and Quantum Circuit (CN114742228B)

Poster Presentations

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems

AQIS 2025 Hong Kong

Aug. 2025

Error Mitigation via N-Representability conditions

TQC 2024 Okinawa and SEEQA 2024 Oxford

Sep. 2024

Tensor-network assisted variational quantum algorithm

AQIS 2023 Seoul

Sep. 2023

Teaching and Mentoring Experience

Peking University (Class Advisor / Undergraduate Cohort Counselor)

EECS Class 10 (Cohort of 2022)

Sep. 2022 – Present

Peking University (Teaching Assistant)

Quantum Information

2023 Spring

Peking University (Teaching Assistant)

Probability and Statistics (A)

2022 Spring

Specialized Skills

Languages: Chinese Mandarin (native speaker), English (fluent)

Programming Languages: Python (mostly used), C++ (familiar), Julia (beginner)